



```
In [1]: import matplotlib.pyplot as plt
import pandas as pd
%matplotlib inline
import numpy as np
```

```
In [2]: customer_data = pd.read_csv('hierarchical-clustering-with-python-and-scikit-learn/customer_data.csv')
```

```
In [3]: customer_data.shape
```

```
Out[3]: (200, 5)
```

```
In [4]: customer_data.head()
```

```
Out[4]:
```

	CustomerID	Genre	Age	Annual Income (k\$)	Spending Score (1-100)
0	1	Male	19	15	39
1	2	Male	21	15	81
2	3	Female	20	16	6
3	4	Female	23	16	77
4	5	Female	31	17	40

```
In [5]: data = customer_data.iloc[:, 3:5].values
```

```
In [11]: data
```

```
Out[11]: array([[ 15, 39],
 [ 15, 81],
 [ 16,  6],
 [ 16, 77],
 [ 17, 40],
 [ 17, 76],
 [ 18,  6],
 [ 18, 94],
 [ 19,  3],
 [ 19, 72],
 [ 19, 14],
 [ 19, 99],
 [ 20, 15],
 [ 20, 77],
 [ 20, 13],
 [ 20, 79],
 [ 21, 35],
 [ 21, 66],
 [ 23, 29],
 [ 23, 98],
 [ 24, 35],
 [ 24, 73],
 [ 25,  5],
 [ 25, 73],
 [ 28, 14],
 [ 28, 82],
 [ 28, 32],
 [ 28, 61],
 [ 29, 31],
 [ 29, 87],
 [ 30,  4],
 [ 30, 73],
 [ 33,  4],
 [ 33, 92],
 [ 33, 14],
 [ 33, 81],
 [ 34, 17],
 [ 34, 73],
 [ 37, 26],
 [ 37, 75],
 [ 38, 35],
 [ 38, 92],
 [ 39, 36],
 [ 39, 61],
 [ 39, 28],
 [ 39, 65],
 [ 40, 55],
 [ 40, 47],
 [ 40, 42],
 [ 40, 42],
 [ 42, 52],
 [ 42, 60],
 [ 43, 54],
 [ 43, 60],
```

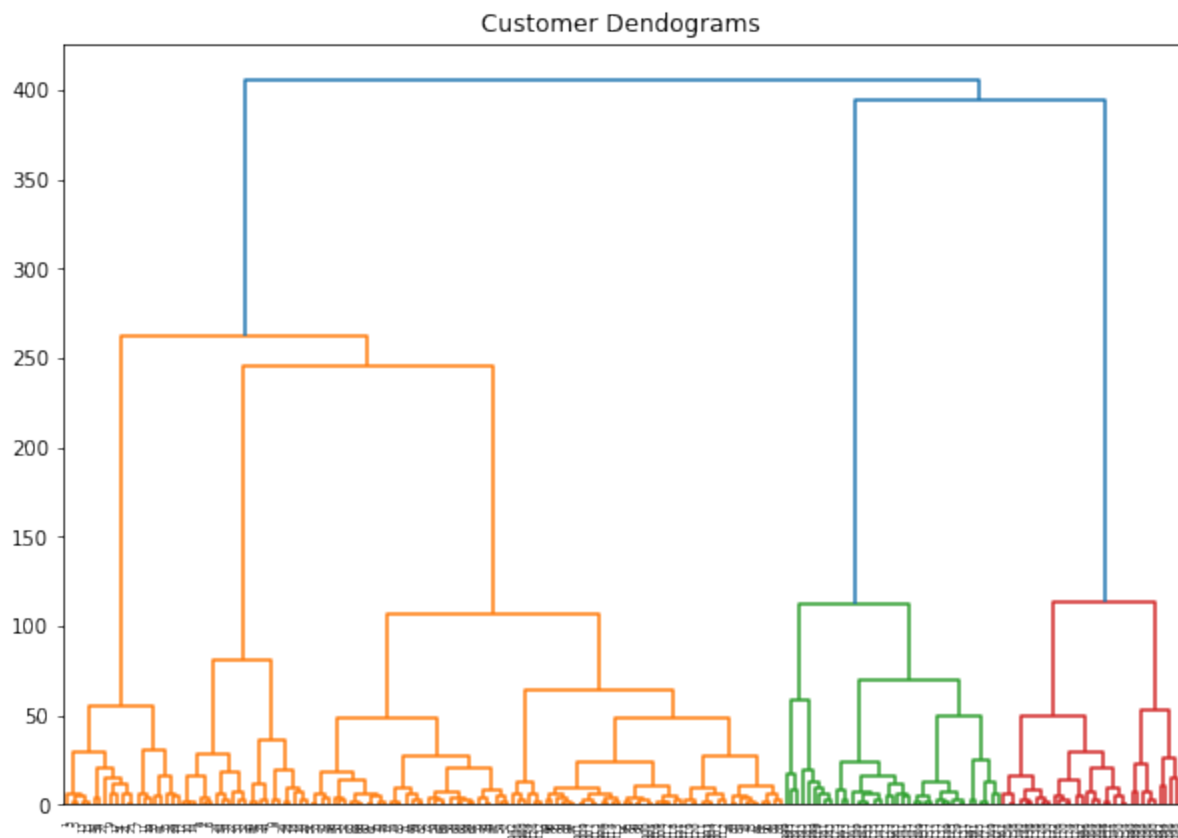
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[44, 46],
[46, 51],
[46, 46],
[46, 56],
[46, 55],
[47, 52],
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[79, 83],

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[ 87, 27],
[ 87, 63],
[ 87, 13],
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[ 88, 15],
[ 88, 69],
[ 93, 14],
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[ 98, 88],
[ 99, 39],
[ 99, 97],
[101, 24],
[101, 68],
[103, 17],
[103, 85],
[103, 23],
[103, 69],
[113,  8],
[113, 91],
[120, 16],
[120, 79],
[126, 28],
[126, 74],
[137, 18],
[137, 83]], dtype=int64)
```

In [6]: `import` scipy.cluster.hierarchy `as` shc

```
plt.figure(figsize=(10, 7))
plt.title("Customer Dendograms")
dend = shc.dendrogram(shc.linkage(data, method='ward'))
```



```
In [8]: from sklearn.cluster import AgglomerativeClustering

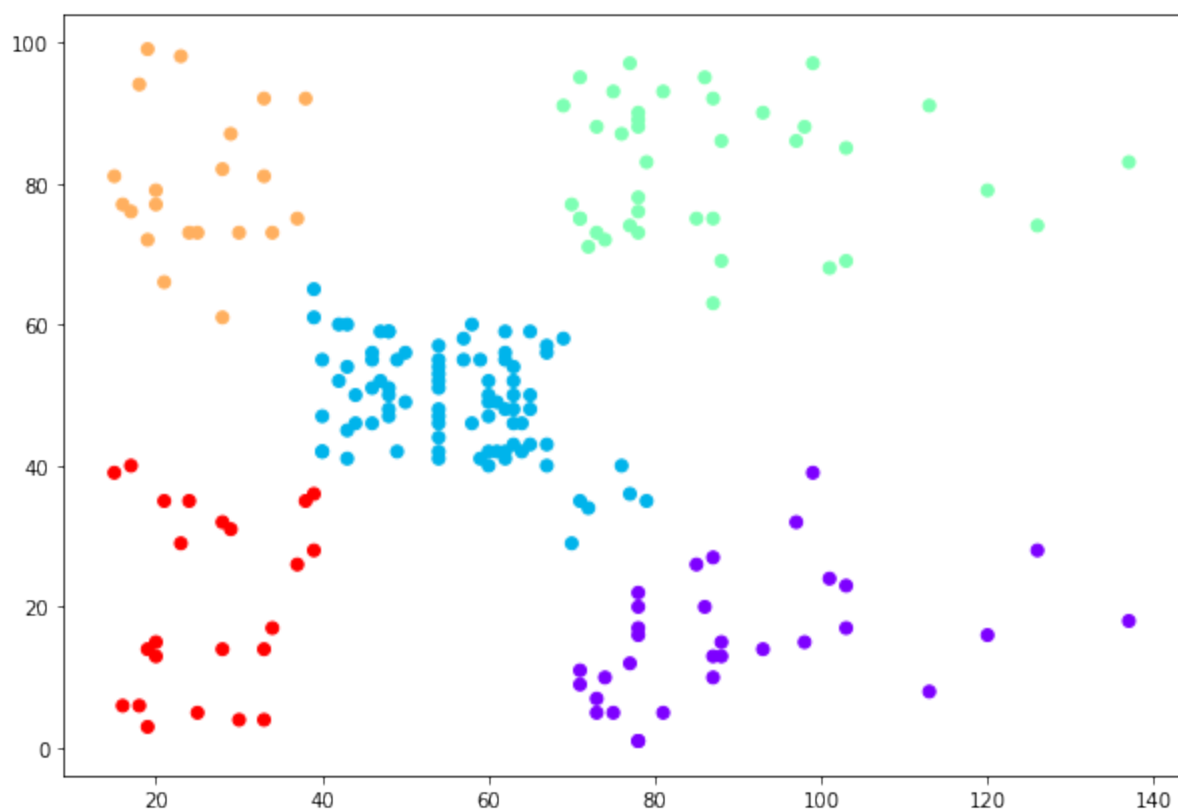
cluster = AgglomerativeClustering(n_clusters=5, affinity='euclidean', linkage=
labels_=cluster.fit_predict(data))
```

```
In [9]: labels_
```

```
Out[9]: array([4, 3, 4, 3, 4, 3, 4, 3, 4, 3, 4, 3, 4, 3, 4, 3, 4, 3, 4, 3, 4, 3,
4, 3, 4, 3, 4, 3, 4, 3, 4, 3, 4, 3, 4, 3, 4, 3, 4, 3, 4, 1,
4, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,
1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,
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1, 2, 0, 2, 0, 2, 0, 2, 0, 2, 1, 2, 0, 2, 1, 2, 0, 2, 0, 2, 0, 2, 0, 2,
0, 2, 0, 2, 0, 2, 1, 2, 0, 2, 0, 2, 0, 2, 0, 2, 0, 2, 0, 2, 0, 2, 0, 2,
0, 2, 0, 2, 0, 2, 0, 2, 0, 2, 0, 2, 0, 2, 0, 2, 0, 2, 0, 2, 0, 2, 0, 2,
0, 2], dtype=int64)
```

```
In [10]: plt.figure(figsize=(10, 7))
plt.scatter(data[:,0], data[:,1], c=cluster.labels_, cmap='rainbow')
```

```
Out[10]: <matplotlib.collections.PathCollection at 0x18f9c15d5e0>
```



In []: